

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce
Department of Information Technology (B.Sc.IT Semester V)
MERN STACK

Practical – IV

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Class: TYIT	Batch:1
Date of Assignment:10-7-26	Date/Time of Submission:15-7-26

4.React Components, Props, and State

a. Create multiple components and render them on a single page.

Code : **Header.jsx**

```
function Header() {  
  return (  
    <div>  
      <h1>Placement Management System</h1>  
      <hr />  
    </div>  
  );  
}
```

export default Header;

Student.jsx function

```
Student() {  
  return (  
    <div>  
      <h2>Student Details</h2>  
      <p>Name : Ashwin Tripathi</p>  
      <p>Roll No : 101</p>  
      <p>Course : B.Sc. IT</p>  
      <p>Semester : VI</p>  
      <hr />  
    </div>  
  );  
}
```

```
export default Student;
```

Eligibility.jsx function

```
Eligibility() {  
  const percentage = 75;  
  
  return (  
    <div>  
      <h2>Eligibility</h2>  
      <p>Percentage : {percentage}%</p>  
      <p>Status : {percentage >= 60 ? "Eligible" : "Not Eligible"}</p>  
    <hr />  
    </div>  
  );  
}
```

```
export default Eligibility;
```

Placementstatus.jsx function

```
Placementstatus() {  
  const placed = true;  
  
  return (  
    <div>  
      <h2>Placement Status</h2>  
      <p>{placed ? "Placed" : "Not Placed"}</p>  
      <hr />  
    </div>  
  );  
}
```

```
export default Placementstatus;
```

Companyoffer.jsx

```
function Companyoffer() {  
  return (  
    <div>  
      <h2>Company Offer</h2>  
      <p>Company : Infosys</p>  
      <p>Package : ₹6 LPA</p>
```

```
    <p>Location : Pune</p>
    <hr />
  </div>
);
}
```

```
export default Companyoffer;
```

Skillprogress.jsx

```
function Skillprogress() {
  const react = 90;  const
  java = 85;
  const python = 95;

  return (
    <div>
      <h2>Skill Progress</h2>

      <p>React : {react}%</p>
      <p>Java : {java}%</p>
      <p>Python : {python}%</p>

      <hr />
    </div>
  );
}
```

```
export default Skillprogress;
```

Footer.jsx function

```
Footer() {
  return (
    <div>
      <p>© 2026 Sheth L.U.J. & Sir M.V. College</p>
    </div>
  );
}
```

```
export default Footer;
```

App.jsx

```
import Header from "../components/Header"; import Student
from "../components/Student"; import Eligibility from
"./components/Eligibility"; import Placementstatus from
"./components/Placementstatus"; import Companyoffer from
"./components/Companyoffer"; import Skillprogress from
"./components/Skillprogress"; import Footer from
"./components/Footer";
import "../App.css";
```

```
function App() {
  return (
    <div className="container">
      <Header />
      <Student />
      <Eligibility />
      <Placementstatus />
      <Companyoffer />
      <Skillprogress />
      <Footer />
    </div>
  );
}
```

```
export default App;
```

App.css

```
.container { width: 600px;
margin: 30px auto; padding:
20px; border: 2px solid
#333; border-radius: 10px;
background-color: #f5f5f5;
font-family: Arial, sans-serif;
}
```

```
h1, h2 {
  text-align: center;
}
```

```
p {
```

```
font-size: 18px;  
}
```

```
hr {  
  margin: 15px 0;  
}
```

Output :

Placement Management System

Student Details
Name : Ashwin Tripathi
Roll No : 101
Course : B.Sc. IT
Semester : VI

Eligibility
Percentage : 75%
Status : Eligible

Placement Status
Placed

Company Offer
Company : Infosys
Package : ₹6 LPA
Location : Pune

Skill Progress
React : 90%
Java : 85%
Python : 95%

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b. Pass data between components using props.

Code :

App.jsx

```
import Header from "../components/Header";
```

```
import Footer from "../components/Footer";
```

```
import HotelCard from "../components/HotelCard";
import "../App.css";
```

```
function App() {
  return (
    <div className="container">
      <Header />
      <HotelCard
        restaurant = "Domino's Pizza"
        food = "Pizza, Pasta, Beverages"
        price = {1599}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "McDonald's"
        food = "Burgers, Fries, Beverages"
        price = {1699}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "Pizza Hut"
        food = "Pizza, Pasta, Beverages"
        price = {599}
        rating = {4.5}
        discount = {20}/>
      <HotelCard
        restaurant = "Chicken Hut"
        food = "chicken Pizza, chicken hande Pasta, Beverages"
        price = {1099}
        rating = {4.5}
        discount = {20}/>
      <Footer />
    </div>
  );
}
```

```
export default App;
```

Header.jsx

```
function Header() {
  return (
    <div>
      <h1>Online Food Delivery Dashboard</h1>
```

```

        <h3>Ashwin</h3>
        <hr />
    </div>
    );
}

export default Header;

HotelCard.jsx
function HotelCard(props) {
    const discount = (props.price * props.discount) / 100;
    const finalPrice = props.price - discount;

    return(
        <div>
            <h2>{props.restaurant}</h2>
            <p>Food Item : {props.food}</p>
            <p>Original Price : ₹ {props.price}</p>
            <p>Discount : {props.discount}%</p>
            <p>Final Price : ₹ {finalPrice}</p>
            <p>Rating : {props.rating}</p>
        </div>
    );
}

export default HotelCard;

Footer.jsx
function Footer() {
    return (
        <div>
            <p>© 2026 Online Food Delivery App</p>
        </div>
    );
}

export default Footer;
Output:

```

Online Food Delivery Dashboard

Ashwin

Food Item : Pizza, Pasta, Beverages

Original Price : ₹1599

Discount : 20%

Final Price : ₹1279.2

Rating : 4.5

Food Item : Burgers, Fries, Beverages

Original Price : ₹1699

Discount : 20%

Final Price : ₹1359.2

Rating : 4.5

Food Item : Pizza, Pasta, Beverages

Original Price : ₹599

Discount : 20%

Final Price : ₹479.2

Rating : 4.5

Food Item : chicken Pizza, chicken hande Pasta, Beverages

Original Price : ₹1099

Discount : 20%

Final Price : ₹879.2

Rating : 4.5

© 2026 Online Food Delivery App

c. Use the useState hook to manage and update data.

Code :

```
import {useState} from 'react';
```

```
function App() {
```

```
  const [count, setCount] = useState(0);
```

```
  return (
```

```
    <div className="App">
```

```
      <h1>Button click Counter App</h1>
```

```
      <h2>User Clicks : {count} times</h2>
```

```
      <button onClick={() => setCount(count + 1)}>Click Me</button>
```

```
      <h3>Ashwin</h3>
```

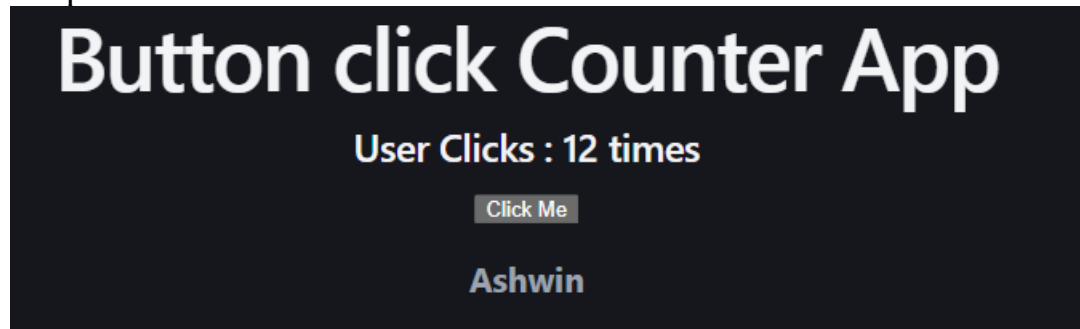
```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

Output :



d. Create a form component to accept user input and display it dynamically.

Code :

Code:

App.jsx

```
import StudentRegisterForm from "../components/StudentRegisterForm";
```

```
function App() {  
  return (  
    <div>  
  
      <StudentRegisterForm />  
    </div>  
  );  
}
```

```
export default App;  
StudentRegisterForm.jsx  
import { useState } from "react";
```

```
function StudentRegisterForm() {  
  const [name, setName] = useState("");  
  const [email, setEmail] = useState("");  
  const [phone, setPhone] = useState("");  
  const [CGPA, setCgpa] = useState("");  
  
  return (  
    <div>  
      <h1>Student Registration Form</h1>  
      <hr/>  
      <label>Name:</label>  
      <input type="text" placeholder="Enter Name" value={name}  
onChange={(e) => setName(e.target.value)} />
```

```

    <br/>
    <label>Email:</label>
    <input type="email" placeholder="Enter Email" value={email}
onChange={(e) => setEmail(e.target.value)} />
    <br/>
    <label>Phone Number:</label>
    <input type="text" placeholder="Enter Phone Number" value={phone}
onChange={(e) => setPhone(e.target.value)} />
    <br/>
    <label>CGPA:</label>
    <input type="text" placeholder="Enter CGPA" value={CGPA}
onChange={(e) => setCgpa(e.target.value)} />
    <br/>
    <button type="submit" onClick={() => {}}>Submit</button>
    <hr/>
    <h2>Student details</h2>
    <p>Name: {name}</p>
    <p>Email: {email}</p>
    <p>Phone Number: {phone}</p>
    <p>CGPA: {CGPA}</p>
  </div>

  );
}

```

export default StudentRegisterForm;

Output :

Student Registration Form

Name:
Email:
Phone Number:
CGPA:

Student details

Name: Ashwin Tripathi
Email: AshwinMvlu@gmail.com
Phone Number: 854542515
CGPA: 9.0

e. Display a list of items stored in the state using map().

Code :

App.jsx

import CompanyList from './components/CompanyList';

import './App.css';

```
function App() {  
  return (  
    <div className="App">  
      <h1>Company List</h1>  
      <CompanyList />  
    </div>  
  );  
}
```

export default App;

CompanyList.jsx

import {useState} from 'react';

```
function CompanyList() {  
  const [companies] = useState([  
    "TCS",  
    "Infosys",  
    "Wipro",  
    "HCL",  
    "Tech Mahindra"  
  ]);  
  
  return (  
    <div>
```

```

    <h2>Placement Companies</h2>
    <h3>Ashwin</h3>
    <hr/>
    <ol>
      {companies.map((company, index) => (
        <li key={index}>{company}</li>
      ))}
    </ol>
  </div>
);
}

```

export default CompanyList;

Output :

Company List	
Placement Companies	
Ashwin	
1.	TCS
2.	Infosys
3.	Wipro
4.	HCL
5.	Tech Mahindra

f. Build a small interactive component (e.g., counter or to-do list).

Code :

```

import {useState} from 'react';
function App() {
  const [count, setCount] = useState(0);
  return (

    <div className="App">

      <h1>React Counter App</h1>
      <h2>User Clicks : {count} times</h2>
      <button onClick={() => setCount(count + 1)}>Increment</button>
      <br/>
      <button onClick={() => setCount(count - 1)}>Decrement</button>
      <br/>
      <button onClick={() => setCount(0)}>Reset</button>
      <hr/>

      <h3>Ashwin</h3>
    </div>

  );
}

```

export default App;

Output :

React Counter App

User Clicks : 4 times

Increment

Decrement

Reset

Ashwin

Code B (To do list):

```
import { useState } from "react";
```

```
function App() {
```

```
  const [task, setTask] = useState("");
```

```
  const [tasks, setTasks] = useState([]);
```

```
  // Add a new task
```

```
  const addTask = () => {
```

```
    if (task.trim() === "") {
```

```
      alert("Please enter a task!");
```

```
      return;
```

```
    }
```

```
    setTasks([...tasks, task]);
```

```
    setTask("");
```

```
  };
```

```
  return (
```

```
    <div style={{ textAlign: "center", marginTop: "30px" }}>
```

```
      <h2>To-Do List</h2>
```

```
      <input
```

```
        type="text"
```

```
        placeholder="Enter a task"
```

```
        value={task}
```

```
        onChange={(e) => setTask(e.target.value)}
```

```
      />
```

```
      <button onClick={addTask} style={{ marginLeft: "10px" }}>
```

```
        Add
```

```
      </button>
```

```
      <hr />
```

```
      <h3>Task List</h3>
```

```
<ol>
  {tasks.map((item, index) => (
    <li key={index}>{item}</li>
  ))}
</ol>
</div>
);
}
```

export default App;

Output :

To-Do List

Add

Task List

1.

running

2.

cooking

3.

time pass

4.

Ashwin tripathi